The Architect and the Machine

How AI was used to build this package — and what that tells you about how I would run it on your projects

Waqas Abbas Mian · Senior ERP Solution Architect (SAP CX → Odoo) — accompanying the MDR Devices Solution Design & Validation Package

1 · Why this note exists

The case brief permitted the use of AI. I want to be precise about what that permission produced, because there are two very different ways to use these tools: as a substitute for judgment, or as a force multiplier under it. This note is the audit trail of the second. Everything in the accompanying package — the live demonstration system, the fifteen documents, the fire-tested compliance gates — was produced by an architect directing AI tooling through a governed loop: **I set direction, the machine produced volume, and nothing survived into the deliverable without verification against the running system.** The tools were not used out of convenience; they were used to guarantee a standard of evidence one person could not otherwise produce against this deadline.

2 · The operating model — dual control, applied to my own toolchain

The main proposal argues that MDR Devices' AI layer must be governed by a simple rule: **AI drafts; it never releases.** I held myself to the same rule. Three disciplines ran through every stage:

- **Direction before generation.** Every workstream began with a written plan — deliverable taxonomy benchmarked against large-consultancy pre-sales norms, scope decisions, the honesty framework (fire-tested / verified live / design-ready / discovery-scoped) — authored as strategy first, then handed to the tooling for production.
- **Verification before acceptance.** No technical claim was accepted from an AI output. Every Odoo capability statement was proven or disproven against a live Odoo Enterprise instance; every regulatory date traced to source (EUDAMED mandate 28 May 2026, legacy deadline 27 November 2026, MDR Articles 18 and 27, GAMP 5). Where the platform contradicted the tooling, the platform won — and the document changed.
- **Adversarial review as a standing habit.** AI-produced work was audited by independent review passes instructed to find faults, not to confirm quality — ending with a word-by-word peer read of all fifteen documents. The defects caught are listed openly in §4: a correction log is evidence of a working quality process, not a weakness.

3 · The journey — five stages, and who did what at each

STAGE	WHAT THE TOOLING PRODUCED	WHAT I DID
1 · Framing & research	Deep research sweeps: MedTech regulatory landscape, Odoo platform capabilities, comparable consultancy deliverable structures.	Set the strategy the research served: that the differentiator would be *honest scoping* — every claim labelled as demonstrated, design-ready, or discovery-scoped — and defined the fifteen-document architecture before a word was drafted.
2 · The live build	Configuration work on a real Odoo Enterprise trial via the platform API: custom compliance fields, automation rules, quality control points, master data.	Decided that "look and feel of the future solution" meant a running system, not slides; specified every control's intended behaviour; when one technical route dead-ended, redirected the build to the API route and kept the deadline.
3 · Multi-agent output, independently audited	Several parallel AI agent runs contributed demo configuration and content at speed.	Commissioned a dedicated audit pass over their combined output rather than trusting volume — it found real defects (§4). Each fix was re-proven by attempting the non-compliant transaction and recording the system's refusal.

4 · Delivery-grade rebuild	Regeneration of the full document set to a delivery standard: consistent house style, embedded live-system evidence, cross-referenced RTM → specification → test → strategy.	Ran forensic consistency sweeps at three levels — document vs. document, document vs. live system, document vs. regulation — plus design and readability passes. Set the acceptance bar; rejected and reworked what missed it.
5 · Peer review before submission	Mechanical sweeps: spelling conventions, encoding artifacts, layout integrity, metadata.	A word-by-word read of all fifteen documents — the discipline a consultancy applies before anything reaches a client. It caught residual internal-draft phrasing and a mixed spelling convention; both fixed, package re-issued. Nothing ships unread.

4 · The correction log — what the machine got wrong, and how I caught it

This table is the point of the note. A tool that is never corrected is a tool that is not being supervised.

DEFECT IN AI-PRODUCED WORK	HOW IT WAS CAUGHT	RESOLUTION & RE-VERIFICATION
A lot's quality-hold date contradicted the process narrative built around it.	Cross-checking every document claim against the live lot record, not the draft text.	Data corrected; the hold gate re-proven by attempting an early release and recording the refusal.
The consignment story was asserted but the consignment location held no stock — an unproven "on-book at the hospital" claim.	A screen-by-screen walk-through of the running system against the pitch narrative.	Serialized unit physically moved to the hospital location; on-book valuation confirmed on the balance sheet view.
The process map sequenced the credential gate *after* consignment dispatch — a control-logic error: a gate that fires after dispatch protects nothing.	Manual review of the diagram against the control design intent.	Re-sequenced to fire at order confirmation, before any dispatch; the block re-tested live against an expired-certification physician.
Costing method silently overrode the intended standard costs, producing wrong valuation figures.	A valuation query returned numbers that did not match the documents.	Category costing corrected; every euro figure in the package re-verified against the ledger.
Compliance fields existed in the data model but were invisible on the record form — configured, but unusable.	Insisting on screenshot evidence for every claim; the screenshot could not be taken.	Fields surfaced in a dedicated compliance section; the evidence now sits in the UAT guide.
Draft documents carried internal working language and an inconsistent spelling convention.	The final word-by-word peer read (§3, stage 5).	Rewritten in client voice; one convention applied across all fifteen documents; package re-issued.

5 · The verification base — why the package can be trusted anyway

The correction log matters because of what stands behind it: **sixteen UAT scenarios executed on the live system** (fifteen passed; one declared simulated rather than disguised); **four compliance gates proven by attempting the non-compliant transaction** and recording each refusal; and every number traceable to a named record in the running instance. A control that has never refused anything is just a setting; every control in this package has refused something, on the record.

> **WHAT THIS MEANS FOR MUCH.**
>
> You allowed AI in this exercise, and I used it the way a modern consultancy should expect it to be used: deliberately, at full leverage, under governance — the dual control principle applied to my own toolchain. The machine provided speed and volume; the direction, verification, corrections and accountability are mine. That is not a compromise between craftsmanship and modern tooling — it is what craftsmanship looks like now, and it is the working method I would bring to your delivery teams.